Part 1

A. Summarize and highlight the ISO risk management standard and their applications in CyberBlast Crispy Company IT security.

ISO 3100 is enterprise risk management made by the iso company it provides a guide for the design, implementation and the maintenance of the risk management process.it helps the company CyberBlast in managing the risk that's encountered by identifying it, analyzing and then evaluating whether the risk should be treated or not (is it a minor risk or major one) in order to satisfy the risk criteria.

Using iso 3100 will help the Cyberblast company to increase the likelihood of achieving objectives of the company and it helps the staff to identify the opportunities and the threats that they might encounter in the future and also enable them to allocate resources to use them for the risk treatment this helps the company to maintain the (CIA) of the company.

Example of the risks: Firewall license expiration

Risk Identification:	ISO-31000 provides a comprehensive approach to identify the obstacles or the threats that might encounter in the future which will prevent us to achieve CyberBlast objectives such as lack of physical security, customer data privacy, authentication and so on. For example, having the firewall expired is a potential risk will lead to unauthorized access and cyber threats because of the lack of security patches (missing) and updates making the network vulnerable
Risk analysis:	ISO-31000 encourages the Cyberblast company to assets the positive and the negative consequence and likelihood to those can occur for having an expired firewall licenses to provide the required input for the risk evaluation and decision making whether risk should be treated or not. CyberBlast company should evaluate the likelihood of having an expired firewall license considering the circumstances and historical data comparing it to the predefined risk criteria to modify a responsive strategy that will prevent a finical loss and keeps the company reputations. The expiration of firewall licenses becomes a real concern as it leads a damage in the company

	reputations causing unauthorized access to sensitive data.
Risk Evaluation:	After knowing what the risk and analyzing its likelihood assessments potential consequence of having an expired firewall CyberBlast company will make the final decision on whether or not to remain the risk is it fall within the acceptable range according to the predefined risk criteria for example having an expired firewall license might be a major risk of its cause a threat to the data security resulting an unauthorized access to sensitive information so if the risk analysis of the firewall resulted an exceeds in the company limits such as finical losses , reputation damage it will result in an major risk.
Risk treatment:	ISO-31000 provides a process for CyberBlast organization that involves modifying the magnitude and the likelihood of the risk consequence to achieve a net growth in benefits. Having an expired firewall license this involve in having measure to reduce the likelihood of the license expiring and it will also help to enhance the organization of the CyberBlast company abilities to respond effecitively.in this way we are not only minimizing the risk but also open up opportunities for growth and improvement by enhancing their overall capabilities by implementing their best practices
Establishing the context:	ISO-31000 encourages the organization to clearly define the scope of the risk encountered and the organization objectives also focusing on the stakeholder expectation and understanding them. In the case of having an expired firewall license this include first to determine which data or the network will be affected then we have to make sure that the efforts to manage the risk align with the CyberBlast primary goal keeping up the a secure IT infrastructure .In addition , meeting the stakeholder's expectation such as the customer expect for their data to be handled securely and not to be accessed by others this will influence our decision focusing no how to maintain the customer trust during the risk.

Monitoring and review:	The steps above are fully reviewed checking if they have not deviated from the plan and making sure that organization private policy also fits the plan.so in the firewall license status a comparison is made with the planned actions whether the renewal or the replacement procedural are applied correctly checking if any deviation occurred also ensuring the security measures are within the organization private policy. Moreover, the monitoring and reviewing process involves taking in feedbacks which contributes future improvements.
Communication and consultation:	This involves understanding the stakeholders' interests (IT staffs , employees)by engaging in an effective communication with the stakeholders and ensuring that the risk management process align within the right priorities (such as emphasizing the importance of having up to date firewall license in minting the network security)offering a clear justification for the decision that is selected such as topics related to the renewal of the firewall license and having a range of risk treatment available options

B & D.

Assets	Control	CIA	Solution
Expired license	Firewall	Confidentiality, Integrity ,Availability	Renew the license of the firewall
Expired Digital certification	VPN	C	Renew the vpn certification
Employees end-POINTS	Anti-malware, HIDS, Cracked version of software	CIA	Deploy HIDS, Update the anti-malware system, and install the original version of the software on employee '
Client data transformation	HTTP, MD5	C, I	Put a secure website HTTPS instead of HTTP

			and use stronger hashing algorithm HMAC
Data backup	Poor Encryption in transmission only (TLS, SSL), SHA1 hashing	C, I	Encrypt the data at rest in the cloud
Email phishing Incident	No control	C, I	Email filtering and Employees training
Inadequate policy	Password length and complexity, hashing for database, data handling policies	CIA	Use a strong hashing algorithm for data base and have a clear data handling polices also make at least 12 long password characters with small & capital letters, symbols, numbers
no isolation between subnets	No control	CIA	Implement DMZ to isolate the servers
Vulnerable OpenSSH Server	OpenSSH	A	Upgrade OpenSSH to a secure version and patch vulnerabilities

C.

Assets	Threat /vulnerabilities	Existing controls	likelihood	consequences	Risk level
Endpoints and Devices	Outdated anti-malware software, cracked version of software installed on the end devices also HDS system isn't available	1. Outdated ani-malware 2. cracked version of software 3.unavailable HIDS (no control used)	High	Major	High
Company Policies and Compliance	1.Inadequate password policy, absence of data handling policies, missing incident response plan, absence of training	1. Inadequate password policy on the employee's device	Almost-certain	catastrophic	Extreme
Third-Party VPN Access and Digital Certificate	Unapproved VPN access from the stakeholders, expired digital certificate that's used for encryption	No Control	Unlikely	Moderate	Medium
Phishing Incident and Email Security	Opening a phishing email with ransomware attachment to it.	No control.	Unlikely	catastrophic	High
Network Segmentation and Firewall	No isolation between servers no DMZ, expired firewall licenses, HR and Finance device are accessible,	No Control	Likely	Major	High
Data Transmission and Storage	Insecure data transmission, MD5 hashing for integrity, unencrypted data in the cloud, HTTP unsecure website	1.MD5 3.HTTP	Possible	Moderate	High

Web Server Security	Vulnerable OpenSSH version, critical CVE with score of 9.8/10	OpenSSH	Likely	Catastrophic	Extreme
Data Center Physical Security	The door was left open and unattended. absence of security cameras. inadequate environmental control, lack of physical barriers, no UPS system	1.The door secured poorly 2.poor environmental control measures	Almost-certain	Doomsday	Extreme

E.

1- Identifying the risk:

In order to identify the risks in the CyberBlast company the first thing the we should do is visualizing the employee's workplace and the activities they go through trying to conducting the risk that they might go through while they are working including the people that they work in the company and the people that don't work in the company such as the maintenance team from other companies. After doing that then we need to dig in the previous risks or incidents that occurred before in the past then after finding all the risks we need to find its likelihood for each one and which organization or people in the company that it might cause harm to. (Andales, 2023) (Anon., n.d.)

2- Decide who might be harmed and how:

After determining the risk, then we started to think about who might get harmed by these affects and countermeasure (the action that we are considering to do) for these risk that might cause in a potential harm for CyberBlast company and A plan that shows who's responsible to take out this action (solving the required risk). For example, not encrypting the data in the cloud that contains the customer privacy will cause harm to the customer also the Company reputation and trustworthy cause the data might get leaked outside causing these threats to occurs in the future. So here we identified who is getting harmed

And how.

3- Evaluating the risk:

After gathering the risk and their impacts on the company we should now priorities them in order in order to do that we should see which of the risk are within the acceptable range of the company and which one is not also how severe the consequences might get if it occurred considering the business goals and the company resources and priorities this will help us creating a risk plan based on their priorities to control this risk within the acceptable range it should be then compare these practices that we applied in order to lower the risk to make it in the acceptable range with the recommended one to determine if any improvement needed or anything needed to be added ,Considering what we already did and what controls we have in the CyberBlast Company.

4- Record your findings and implement them:

Recording the findings that we found I crucial to keep the company stakeholders updated and the organization of the company making sure that they have a clear understanding for what's going on and the plans that we conducted to do and the countermeasures. The thing that should be included in the record is the step above:1. identifying the risk,2. creating the likelihood for these risk with a document that include the harmed parties and how. 3.Risk evaluation results and the priority plan that was created

for these risks based on the company goals and resources Also Identifying the controls that were used 4. Countermeasures plan this involves in the action needed to mitigate this risk 4. then identifies the responsibility for each employee or organization their roles and responsibility for implementing the countermeasures needed to eliminate the risks for example this include giving the employees their roles on this risk such as the taking form the security regular updates about the security controls. 4.The timeline table for example how long does it take to eliminate this risk or to lower them within the acceptable range .5. Improvements which include compare these practices that we applied in order to lower the risk to make it in the acceptable range with the recommended one.

5- Review your risk assessment and update if necessary:

The workplace might change overtime so as the risk that might occurred to the CyberBlast company. Implementing new technologies, new employees are introduced to the company as things changes that probability of having a new risk increases in order to avoid this a continues review on the risks and updating the process help us to overcome these challenges and stay on the top. Such as regular reporting to the stake holders about the risk's updates and continuous monitoring and training ensuing that everyone is aware of his role these steps help us to stay on the top.

F. Explain data protection processes in details and regulations that can enhance CyberBlast Crispy IT data security.

(Anon., n.d.) (Anon., n.d.)

What Is Data Protection and Why Is It Important?

Data protection is protecting the data from getting comprised or getting corrupted from any unusual activity also providing the capability to restore the data to it original form if anything happened the main There is certain measure to protect the data, First point is having an CDP which backup any latest update happened into the data helping in minimizing the data loss CyberBlast company implanted CPD to make sure the data was secured and if any loss to data happened we can restore it ,The second point is limiting the access to any sensitive data for CyberBlast company or for customer information, CyberBlast implanted this by having different access privileges used by the employee ensuring that the data won't be face any kind of danger , The third point is that the CyberBlast decides to implement encryption for the data during the transmission process to secure it if any of data was hacked ,corrupted in the process it wont be understandable ,The fourth point the CyberBlast company decided to conduct security training and awareness, which will helps the employee to make the best practices in the upcoming threat that might occur rising their awareness about it helping in creating a secure environment

Part 2

Part2: Security Policy

A. Design a suitable security policy for CyberBlast Crispy include and highlight the main components of the CyberBlast Crispy disaster recovery plan. (it should be included in the document as an appendix using SANS policy forms). (IN THE APPEDNIX)

B. Develop security plans (derived from the designed policy) for the top three critical assets and justify the selected elements.

1. Password Policy: -

In Order to have a strong and robust password there is some criteria that's The CyberBlast followed. First, the password should be at least 12 letters long and the letters would be mixed for example lowercase letters, symbols, and number all of these should be add in the 12 letters. Secondly, the password should be changed regular bases on a specific time for example every month. Thirdly, the password that was created from CyberBlast should be only used for the company its prohibited to used it to anything unrelated to work for example personals accounts. Fourthly, its prohibited to save the password on any outside platform on the internet making sure that "Remember password" feature is not selected.

2. **Email Policy:** -
   In Order to follow the email policy, the emails should be used only for related business works anything outside the business works for example personal emails is prohibited. In Addition, the emails used in CyberBlast company should be within the policies and the procedure of compliance of the company. Also, any email from CyberBlast company cannot be used in any kind of creating offensive message such as race, gender, religious beliefs if any employees receive these emails that should be reported directly.

3. Anti-Virus Software Policy: -

In Order to have a follow the anti-virus software there is some points need to be followed Firstly, all devices from CyberBlast company their anti-viruses software should be updated regularly and checked on regular bases if its working effectively "is it preventing malware threats?". Secondly, employee is not allowed to disable any anti-virus protection from their devices. Thirdly, all employees devices will have a scanning plan for example every week the IT security teams will scan the devices to look for anything suspicious this will ensure the integrity of CyberBlast information system.

C. Evaluate the suitability of the tools used in this policy.

- **Firewall:** firewall will control the incoming and outgoing traffic from the network acting as a barrier between the external up coming threats and the internal network also prevent unauthorized access for this to happen, they should be a regular updating of license and the configuration to make sure that that's its working efficiently eliminating any potential risk exploited by the attackers.
- **Antivirus Software:** it helps us to protect any ends point in CyberBlast organization against the malware for this to happen the antivirus should be up to date to recognize all malware type and preform a regular scan on the endpoints devices to ensure their effectively by detecting, removing and preventing malwares.
- **Encryption Tools:** using the most robust and complexed algorithm will convert the data onto unreadable format and can be accessed only by the decryption key its essential for the CyberBlast company to be implemented in the data that's in rest and in transmit to ensure the integrity of the data and confidentiality and even if unauthorized access occurred it won't be readable this helps in securing sensitive data.

D. Discuss, in detail, the security impact of any misalignment of IT security with CyberBlast Crispy policy and recommend how to maintain their alignment. (Brandenburg, 2023)

Security impact of any misalignment of IT security with organizational policy:

Increased Vulnerability:

Misalignment between the its security and the CyberBlast company this will push the IT Security to not to follow the CyberBlast objectives causing in ignorance or unnoticing the risk that might occur this will result the critical data to be left vulnerable making CyberBlast company more susceptible to get attacked.

Non-compliance with Regulations:

Not following the CyberBlast policies my result in non-compliance which result the IT security not following the standard of the company and the regulation this will expose the company to have legal consequences and fines because the sensitive of the customer information will be exposed resulting in damaging the CyberBlast reputations and finical loss.

Data Integrity and Confidentiality Risks:

Unauthorized access CyberBlast company's sensitive data might occur cause the IT security team is not following the company policy will get this is not only affects the CyberBlast company but also effects customer and stakeholder trust as their information is not handled in established security protocols

Recommendation on how to maintain their alignment:

1.Employee Training and Awareness:

Training the employee and the staff by doing set of activities helping them to follow set of rules and improving their working skills will help the employee to have a good relationship with each other and decrease the work conflict since the training will enhance their working skill level. Rasing the employee awareness will help them to identify their role in the company and how does it effect in maintaining a secure environment

2.Collaboration Between IT Security and Business Leaders:

Conducting a regular meeting between the IT security and business leader will help to enhance in achieving the CyberBlast goals, priorities and challenges that might occur making sure that the decision is made from different point of view and everyone interacted in the meeting such as giving feedback and opinions about what's going on and this will help to achieve a strong relationship between the IT security and business leaders helping in achieving CyberBlast policy goals

3.Continuous Monitoring:

Conduct a Regular monitoring about the behavior of the employees and the work practice they are doing does it match the CyberBlast also reviewing if the employees can access only the authorized metrical for them based on their role and responsibility.

Appendix

CyberBlast Crispy Security Policy

Policy Components:

1. Information System Security Policy (ISSP):

Policy that identifies certain risks and threats and how to target them to protect the CyberBlast and what needed to be implemented to protect the CyberBlast from the upcoming threat. Sample were used in ISSP

- Password policy: it's a set of rules that created based on the CyberBlast policy goals to enhance the security awareness encouraging the employee to use strong password to prevent unauthorized access to our sensitive data. All staff that have the access to CyberBlast company system are responsible to the appropriate measurement such as changing it frequently or limit the password attempts.
- Email policy: Electronic email is used for the staff to communicates within the organization, but the misuse of email can become a security risk therefore it's important to raise the awareness for the users to give them the appropriate policy on how to use it what acceptable and what not acceptable also including policies will raise the employees experience on recognizing email-based threat like phishing

2. Enterprise Information Security Policy (EISP): -

EISP is owned by the C-level, it helps in writing a specific policy with identifying the scope of the company and the responsible organization for this policy also components with the companies frame work

- Sensitive information handling policy: This helps us to maintain the confidentially and the integrity of handling sensitive information. CyberBlast policy provides a clear understanding on the procedures that are required to implement in order to handle sensitive data such as establishing encryption and a secure transmission for these data

3. System-Specific Security Policy (SSP): -

SSP its specialized for the equipment and it might differ on some of the devices for example access control list, router configuration, restricting the user accessibility such as printers. Sample were used

- Anti-virus software policy: The usage of Anto-virus is all over the system in order to mitigate any malicious malware threats or attack on the system. We enforce the installation of Ant-virus software and regular checking if regular scanning is procced and are they up to date these ensure that Anti-Virus is working effectively. This policy helps in preventing damage to the user applications, data, files, and hardware.

4. Aligning DRP with Security Policy:

This help the CyberBlast to ensure that the disaster is handled within the company policy rules and regulation these are the certain objectives that include DRP.

- **4.1. DRP Team:**

  When a disaster occurs, such as data loss or power outages, a recovery plan comes up to solve this disaster as quickly as possible and efficiently to do this we have to create a recovery plan the first step to do this is have a disaster team from the company with different expertise working in different areas this might include CIO, IT specialist and maybe the stakeholder of the company this help the CyberBlast organization to identify and mitigate the risks that are associated with the recovery disaster plan.

- **4.2. Recovery Time Objective (RTO):**

  RTO means how much time does the drp team need to solve the disaster for the application to be available again. Cyberblast drp put down a logical time to solve the power outage that occurred and solved it in 30 minutes which is incredible we prevented to have more data loss also prevented the disaster to affect the CyberBlast reputations.

- **4.3. Recovery Point Objective (RPO):**

  It's the amount of the data that CyberBlast company lost during the disaster this depends on the RTO. If we had a small time to solve the disaster, a small amount of data will be lost. So, the better the RTO, the better the RPO will be. The RPO process can help us maintain the company regulatory requirements.

- **4.4. Backup Strategy:**

  Having backups is essential for the company. It helps in reducing the time it takes to recover from the disaster with the least loss. In order to achieve this, backup should be kept in safe places that can outstand fire and water or anything that might affect it and it should be tested regularly if we did that then we can use our backup as a resource if all the data got lost maintaining the availability during the CyberBlast disaster.

- **4.5. Documentation:**

The disaster should be documented including al the procedures such as data backup, communication plan to keep all the stakeholder to keep them informed, resources and the equipment needed, and etc., that DRP team went through or the steps that they want to follow in order to recover from the disaster it provides the team a road map on how thing will be executed and in which order. Therefore, not having a documentation might cause the team not following it in the right order or the recover might not be that efficient. The documentation might also be updated and reviewed afterward if anything needs to be added in order to keep up with the Cyberblast company regulation and structure
this helps the documents to stay up to date and effective.

- **4.6. Automation**

  Automation enables the team in the CyberBlast company to respond faster in the plan recovery disaster as well as ensure that each procedure or step is taken out in the right order and consistently helping to minimize the data low that can occur during the disaster recovery process. Using Automation will enhance the efficiency, human error, reliability, and accuracy of the procedures this will make the recovery faster.

Part 3

A. Analyze the IT security audit and its impact on CyberBlast Crispy IT security.

Analyze the IT security audit on CyberBlast Crispy IT security.

Me as an IT auditor I was searching for anything vulnerable across the CyberBlast information system such as physical components of CyberBlast information system and the environment in which the information system is housed in for example the environmental control system CyberBlast company were outside the acceptable range based on the policies and regulation of the company also they had weak physical security protection used for the data center such as old key metal and no security cameras were found so I recommended the CyberBlast organization to implement Security Cams ,Biometric access ,improve environmental controls. Secondly, we looked for the applications and software used, we noticed that there was vulnerable OpenSSH version and outdated software system were used result in a lack of security patches causing threat to the data integrity of the company. Thirdly, there was some Network vulnerabilities problem such as expired firewall license and no isolation between different subnets increasing the threat of having an unauthorized access and insecure data transmission so I recommended to implement DMZ to separate the subnets and a vpn channel to secure the data transmission process. Lastly, we noticed

that there was incorrect human dimension, including how there was an insufficient password policy, absence of data handling policy, and the phishing risk. My recommendation for this is for the CyberBlast company to implement the suggested measurements by conducting a regular security audit to follow and to establish a strong password policy, phishing awareness, and data handling following these approaches will help the CyberBlast company to have a strong security for their information system.

What IT Audits Effects on CyberBlast Company

1.Ensures that data is protected:

IT security auditor helps in establishing a regular bias auditing routine covering elements such as encryption practices, network access control, data transmission, and etc.
This routine will ensure the safeguarding of the sensitive data.

2. Observe operations from different perspectives:

Through examination we were having we had sight from different areas in the company will provides an more knowledge not only specified to only cybersecurity area. this wide view will allow us to solve problem based on the overall CyberBlast goals

3.Point out security flaws in system:

IT auditor will examine CyberBlast cybersecurity measures and documenting the flaws that were faced this plays a crucial when search for a solution in CyberBlast company to write down and list what issues we were facing and pointing out the weakness and flaws form the beginning will help the it auditor to give out the best approach and the most secured on to keep the system secure.

B. Discuss the roles of CyberBlast Crispy stakeholders in implementing security audits.

Management (C-level): the management is the one who will make sure that IT auditor recommendation was implemented and every recommendation were took into

consideration. In addition, C-level they will make sure that the resources are being used within the recommendation also taking into consideration that the implementation aren't going to affect the CyberBlast operation.

IT Officers:

Checking every it auditor recommendation for the technical security measurement were implemented and focusing that the implementation was implemented correctly without effecting the CyberBlast performance.

Risk Owners:

Risk owners they identify the risks and assess the potential risk the CyberBlast company might face they usually work with the c-level and the IT officers to mitigate the risk which was identified they tend to make sure that the risks are associated with the security vulnerabilities and they are within the acceptable measurements.

Facility and security officers:

They mainly focus on the physical security of CyberBlast's premises and data center such as implementing security cams and other stuff that was recommended by it auditor and they make sure that the recommended security measure from the IT auditor were implemented in the right place and the data center is kept secured with the following security measurement.

Compliance officers (or Quality assurance):

Ensure that the whole CyberBlast organization implemented every recommendation from the IT auditor making sure that the implementation is relevant to the company's regulation and policies and it does meet the qualifications and the standard.

Part 4
PowerPoints

1.Provide and discuss the top 5 possible risks within the CyberBlast Crispy system, including their likelihood, potential exploitation consequences, risk levels, and suggested controls (countermeasures).

Risk 1: Endpoints and Devices

ASSET: Outdated anti-malware software, cracked version of software installed on the end devices also HDS system isn't available

EXISTING CONTROLS:
1.Outdated ani-malware.
2. cracked version of software.
3.unavailable HIDS (no control used).

LIKELIHOOD: High.

CONSEQUANCES: Major.
RISK LEVEL: High.

COUNTERMEASURES:
1.install original software version
2.setup HIDS
3.update anti-malware

Risk 2: Company Policies and Compliance

ASSET:
Inadequate password policy, absence of data handling policies, missing incident response plan, absence of training

EXISTING CONTROLS:
1. Inadequate password policy on the employee's device

LIKELIHOOD: Almost-certain.
CONSEQUANCES: Catastrophic.
RISK LEVEL: Extreme.
COUNTERMEASURES:
1. implement a strong password policy
2. implement data handling policy

3. create an incident response plan
4. create a regular training schedule

Risk 3: Phishing Incident and Email Security

ASSET: Opening a phishing email with ransomware attachment to it.
EXISTING CONTROLS:
No control used.

LIKELIHOOD: Unlikely.
CONSEQUANCES: Catastrophic.
RISK LEVEL: High.
COUNTERMEASURES:
1.implement email filtering and anti-phishing tools also conduct a regular training on recognizing phishing emails
2. implement software that detect ransomware attachments.

Risk 4: Network Segmentation and Firewall

ASSET:

No isolation between servers no DMZ, expired firewall licenses, HR and Finance device are accessible**,**

EXISTING CONTROLS:

No controls only firewall

LIKELIHOOD: Likely**.**
CONSEQUANCES: Major.

RISK LEVEL: High.

COUNTERMEASURES:
1. Setup DMZ
2. Renew firewall license
3. Restrict access to HR and finance devices.

Risk 5: Data Center Infrastructure

ASSET: The door was left open and unattended. absence of security cameras, Inadequate environmental control, lack of physical barriers, no UPS system
EXISTING CONTROLS:
1.The door secured poorly
2.poor environmental control measures
LIKELIHOOD: Almost-certain.
CONSEQUANCES: Doomsday.
RISK LEVEL: Extreme
COUNTERMEASURES:

1. Implement robust environmental controls
2. Implement enhancement for door security
3. Setup UPS
4. Implement strong physical barriers.

2.Examine the risks associated with and potential consequences of improperly configuring network security devices, including firewalls and third-party VPNs, in the context of IT security

1.Potential consequences of improperly configuring firewalls:

Improper configuration of the firewall will put the CyberBlast Crispy at risk such as unauthorized access causing the hacker to see the sensitive data and exposure it to more threats also it might also accept some of the legitimates threats traffic to reach the company devices decreasing the productivity of the company making user to face bad experience also allowing the traffic to pass will make it more suspectable to be attacked by cyberattacks and increasing in damaging the company reputation because there is no data integrity for the sensitive data which leading us to face some legal consequences and finical fines.

2.Potential consequences of improperly configuring third-party VPNs

Improper configuration of the third-party vpn my lead to unauthorized access to the network or it might also expose some sensitive data such as customer credential information might got leaked causing anyone to gain access to this information or he might use it to gain access to other data breach and this might cause CyberBlast to face finical fine and legal consequences also its also cause non compliance which mean that if the data is not transmitted securely so this is fails in protecting the data which might lead in damaging the data reputation.

3.Examine the enhancement of CyberBlast Crispy's network security through the implementation of network security techniques, including DMZ (Demilitarized Zone), static IP (Internet Protocol), and NAT (Network Address Translation). Provide a comprehensive analysis of how each of these techniques contributes to the bolstering of network security, offering detailed explanations for their effectiveness.

DMZ (Demilitarized Zone:

DMZ its act as an extra layer protection between the local area network and the unauthorized access to prevents any external threats this help in maintaining secure communication with the external networks while protecting the internal systems and DMZ server reduce the risk of getting the Company server attacked these helps in maintaining CyberBlast Crispy’s network infrastructure with these features.

static IP (Internet Protocol):

Static IP helps in assessing Ip address to all the devices in the networks this helps in reducing IP conflicts which ensure stability in network configuration cause static IP doesn't change so this helps in reducing the risks of addressing conflicts and having an unchangeable Ip address and predictable one will help in creating a more secure and stable network environment. In addition, having a static Ip address will help in efficient network management and a smooth interaction because static IPs facilitate a clear identification of the devices and straightforward at tracking them,

NAT (Network Address Translation):

Nat improves the security by acting as a barrier between the private network and the public on it prevent the use of public network it translate multiple private network internal devices IP addresses to one public Ip address that can be seen outside the private network , reducing the risk for external users to access the internal devices directly so this helps to hide the internal infrastructure of the network so what we understand is the Nat act as extra layer of security which helps in reducing the exposure of internal network infrastructure and reducing the number of ip4 public addresses being used.

4.Analyze the benefits and justification of using a Network Monitoring System.
(EXTERNETWORKS, n.d.)

Detect Intrusions in Real-Time
By monitoring our networks, it helps us to detect real-time intrusions for an example when a threat occur you can immediately stop it before it affects anything in the system by preventing the malicious activities, it also can help us in identifying where did the threat come from what are the things that we need to do to stop this risk in the future.

Improve Efficiency and Productivity

Seeing everything how the traffic flow through the network monitoring each device activities checking its functionalities and how the resources are being used and where

Protect your organization from Cyber Attacks
Using monitoring tools will help in searching for the weakness and identify them that cyberattacks might go through to improve it making it harder for the cyberattack to gain access to the CyberBlast information system

5. Evaluate a range of physical and virtual security measures that can be implemented to safeguard the integrity of IT security, considering a minimum of three measures for each category.

Physical security measure:

Physical security refers to protection of the company building, equipment, and data center everything that contains the company information so in order to protect these from any threat there is some measure need to be followed.

Let's list three measures of physical security:

1. Surveillance Cameras

Providing a security cameras and access control system in the places that was recommended from the CyberBlast company the cameras should get checked up regularly so we can keep track of the security of the building did anyone broke the rules or is their any threat to the data center if any accident occurred their would-be recording tape save in a secured place, we need to make sure that the disposal, placement, and storage of cameras tape are secured.

2 Monitoring and alarm, policy, and procedures:
They type of monitoring system or alarm will be decided form the security plan of the company. Focusing on the computer system, entrances, power room, and parking space these help in any unauthorized people to detected and prevented from entering ensuring that an effective way to protect CyberBlast providing an extra layer of security also the security plan limited employee who has the access to turn off by bass system.

3.Biometric Access Controls (Fingerprint, Retina Scan):
Having a biometric in the access control is a good security cause biometric cant be stolen and its unique for each person. The use of Biometric security will enhance the overall security measures offering a reliable way of identifying the authorized user helping us in safeguarding the data to protect CyberBlast organization

Virtual security measure:

Let’s list three measures of virtual security:

1 Multi-Factor Authentication (MFA): Having an MFA will help us to guarantee the security of the system by adding extra layer of security requiring the customer to do two steps of authentication to access the data for example the first one is entering their password and the second one is entering some authentication code sent to their mobile.

2. Anti-malware: having anti-malware on CyberBlast endpoints devices will prevent any malicious software to take over the sensitive data of the company it done this by scanning, detecting, quarantine, and eliminating any malicious code or behavior, maintain the security of the data.

3 Encryption:
Its changing the data into unreadable form to guarantee that no one can steal the data or read it except for the authorized users using the decryption key in this way it maintains the integrity because it would not affect the company if the data were accessed